

WaterShade

The key factors to preserve thermal quality of forested streams and rivers

Literature review – January 2025

Summary

Thermal quality of river networks are conditioned by complex thermal exchanges between air, soil and groundwater as well as by direct insolation to surface waters.

Forestry harvest activities, and especially clearcutting, induces a reduction of transpiration and thermal exchanges at the soil level, especially solar radiation, which contributes to the warming of the soil as well as warmer runoff water and groundwater before they reach streams.

Riparian vegetation and associated shading help maintain or cool water temperatures of the river by protecting water from direct solar radiation. It is a particularly efficient method for headwater streams, with a thermal effect on the entire downstream waterways.

Consequently, the best way to protect or improve the thermal quality of a forested river network in a changing climate context is to combine:

- 1) the protection of headwaters and thermally sensitive areas including wetlands, intermittent streams, saturated soil and shallow groundwater areas from intensive logging, especially clearcutting.
- 2) the preservation of shading buffer zones along permanent streams, with a minimal width of 10 m of tall and dense vegetation.
- 3) a specific protection of the coldest portions of the river network (i.e., the most important existing cold-water patches) which are of crucial importance as thermal heterogeneity is necessary for cold-water fishes residing in relatively warm river networks.

These recommendations are specific to water temperature preservation and other considerations may apply when considering wider ecological integrity (e.g. wider buffer zones).

Introduction

The thermal regime of a watercourse is determined by the following factors (Caissie, 2006; Johnson & Jones, 2000):

- direct insolation of the freely flowing water.
- thermal exchanges by conduction at the water-substrate interface of the watercourse and water-soil interface on the banks.

- thermal exchanges by advection from groundwater inputs and tributaries.
- thermal exchanges by evaporation and convection at the air-water interface.

Consequently, every event or process that affects one of these factors at some place in a watershed may have an impact on local and downstream water temperature. This can be a natural phenomenon such as forest fires, or anthropogenic such as human activities like forest harvesting, road networks, mining, and climate change. In the case of forest management activities and especially forest harvest, warming is driven by two processes: (i) the increase of thermal exchange at the soil level and (ii) the decrease of shading (and thus increase of direct insolation) along water bodies (e.g., Poole & Berman, 2001).

1. Headwaters

Headwaters include wetlands outside of floodplains, small stream tributaries with permanent flow, tributaries with intermittent flow (e.g., periodic or seasonal flows supported by groundwater or precipitation), or tributaries or areas of the landscape with ephemeral flows. These headwaters perform ecological functions (i.e., biological, geochemical, and physical processes that occur within an ecosystem) that are critical for ecosystem services throughout their drainage basins (Colvin *et al.*, 2019). Even intermittent streams can generate a significant source of cold water at their confluence, although there is no surface water flow in summer, due to hyporheic flow (Ebersole *et al.*, 2015).

Regarding water temperature, headwaters play an important role in the thermal regime of a river system and its response to land use and forest harvesting (Hawes & Smith, 2005; Yearsley *et al.*, 2019; Cunningham *et al.*, 2023). First, these streams are highly sensitive to surface and subsurface processes. In many timber harvest studies, the smallest streams experienced the greatest increase in stream temperature following forest removal (e.g., Brown and Krygier, 1970). Secondly, the warming of water in these small streams has a downstream effect on larger streams and rivers (Erdozain *et al.*, 2022), which may mitigate the efficiency of buffer zone protection for these larger watercourses, especially in early summer (Bourque & Pomeroy, 2001; Johnson, 2004).

In such headwater streams and drainage areas, the quality and resilience of summer cold-water habitat is generally regulated by stream discharge, longitudinal stream channel connectivity and groundwater exchange. The removal of forest cover induced by logging, especially clearcutting, increases insolation of exposed ground, reduces transpiration and increases thermal exchanges at soil level, especially solar radiation, which contributes to the warming of soil as well as runoff water and groundwater before they reach streams (e.g., Bladon *et al.*, 2018; Cunningham *et al.*, 2023; Dan Moore *et al.*, 2005; Erdozain *et al.*, 2022; Hartman et Scrivener, 1990; Ice, 1999; St-Hilaire *et al.*, 2000).

However, many factors affect water temperature in streams and the way it will be affected by logging in a given drainage area:

- The logging area: some studies have found a correlation between the logging area within a watershed and the water temperature increase in streams (e.g., Cunningham *et al.*, 2023), especially when the watershed is underlain by resistant geology (Bladon *et al.*, 2018).
- The type and intensity of logging: the effect on water temperature has been shown to be more significant when logging is clearcutting than when it is partial cutting, even when clearcutting is done in the context of a lower overall intensity of forest activity (Erdozain *et al.*, 2022, see below).

- The hydraulic gradient (slope): the watershed slope can be positively or negatively correlated with stream water temperature depending on elevation, geology and groundwater depth (O'Sullivan *et al.*, 2019). A low gradient was found to be linked to a higher warming rate of water after logging by Bourque & Pomeroy (2001).
 - The catchment-to-sun orientation (aspect): south-facing lands and hillslopes get more solar radiation and therefore more warming potential (Bourque & Pomeroy, 2001; Cunningham *et al.*, 2023; Smith & Redding, 2012).
 - The density of wetlands: wetlands can have a mitigation effect on water temperatures in streams depending on their size and location in the watershed (Mellina *et al.*, 2002; Monk *et al.*, 2013). Janish *et al.* (2012) found that stream temperature after logging was higher when the timber harvest includes larger areas of exposed water surface area (streams and wetlands) and saturated soils upstream of monitoring stations.
 - The catchment lithology: a permeable and fractured bedrock may induce a buffering effect and greater groundwater exchanges in summer periods, leading to the mitigation of the effect of forest cutting on water temperature as compared to catchments underlain by resistant geology (Bladon *et al.*, 2018; Monk *et al.*, 2013).
 - The soil texture: coarse textured deposits are permeable, which is usually associated with groundwater discharge or hyporheic processes in rivers. Groundwater in such sandy soil is more sensitive to surface warming than in clayey soil and peat soil (Kurylyk *et al.*, 2015).
 - The groundwater depth and recharge rate. The thermal regimes of certain headwater streams or smaller rivers are closely influenced by streambed advective heat fluxes induced by groundwater–surface water interactions (Caissie, 2006; Kurylyk *et al.*, 2015). Shallow aquifers are sensitive to climate change, permanent deforestation, and wildfires, leading to warming groundwater temperature and, consequently, warming stream water temperature. In some cases, shallow groundwater warming may eventually exceed the magnitude of surface water warming (Kurylyk *et al.*, 2013).
 - The original forest composition: the greater amount of mixed forest in the drainage area has been found to promote cooler median temperatures in confluence plume tributaries, more than other types of forest (Monk *et al.*, 2013).
 - The distance from logging area: depending on land use in other parts of the watershed, the effect of logging may be mitigated downstream by cool water inflows from other tributaries and/or groundwater that are protected or unimpaired.

Some of these studies were carried out specifically in New Brunswick:

- Bourque and Pomeroy (2001) compared water temperature in streams of five small watersheds in southern New Brunswick, taking into account historical forest harvest in each one. They found that the increase of summer water temperature was a function of total insolation and level of deforestation in the watershed.
- Alexander *et al.* (2003) measured shallow and deep groundwater temperature in an upslope clearcut area and in the downslope buffer strip of the Catamaran Brook (tributary of the Little Southwest Miramichi River watershed). For the period studied (August 2001 to October 2002), temperature of ground surface, shallow groundwater and deep groundwater in the streamside buffer strip was more stable and cooler than in the clearcut.
- Erdozain *et al.* (2022) conducted comparative studies across three forest watersheds of varying management (intensive, extensive and minimal) in northern New Brunswick between 2014 and

2018, considering several water quality indicators including water temperature. They found that water temperature in small streams was warmer in managed watersheds, especially in the one with extensive harvest, than in the one with minimal harvest. They concluded that the effect of forest harvest on water temperature and other water quality indicators is more significant when logging is clearcutting than when it is partial cutting, even when clearcutting is done in the context of a lower overall intensity of forest activity.

All this information shows that the effect of forest harvesting on water temperature is multifactorial and highlights the need:

- (i) to better regulate and distribute harvesting in time, space and intensity at the watershed scale, especially in the upper catchment of streams which is appealing for intensive harvesting and clearcutting due to the low surface gradients and lack of visible surface water
- (ii) to avoid clearcutting as far as possible
- (iii) to adapt forest harvest practices to watershed characteristics such as geology, groundwater, wetlands, topography, soil texture and aspect.

2. Buffer zones

The buffer zone (or buffer strip or riparian buffer) of a watercourse provides a progressive transition between the stream and the surrounding area (forest, agriculture) which influences some characteristics of the river ecosystem such as water and air temperature, physical and chemical processes in the ground litter, as well as habitat of semi-aquatic and hydrophilic species (Chen *et al.*, 1999). Buffers also provide organic material (leaves and woody debris) to watercourses and wetlands that is a source of nutrients and habitat structures for animals. They also promote infiltration of runoff water and filtration of sediments originating from surrounding areas with exposed soil, before reaching the watercourse.

The trees and shrubs in the buffer zone help maintain or cool the thermal regime of the river by protecting water from direct solar radiation. It is particularly efficient for narrow headwater streams (e.g., Jackson *et al.*, 2021). The effect of riparian buffers on water temperature depends on several factors such as channel geomorphology, aspect, gradient, streambed substrate, streambed-subsurface exchanges of heat and water, presence of stream-adjacent wetlands, as well as vegetation density, type, and height (e.g. Cunningham *et al.*, 2023; Dugdale *et al.*, 2018; Gomi *et al.*, 2006; Jackson *et al.*, 2021; Poole & Berman, 2001).

The removal of buffer zones has a strong impact on the stream thermal regime by reducing shading and increasing solar radiation on water. These impacts include:

- a faster increase of temperature in early summer (Johnson & Jones, 2000)
- an increase of mean water temperature during summer, up to +2-3 °C in some studies, as compared to streams with coniferous buffer zones (Brown *et al.*, 2010; Webb and Crisp, 2006)
- an increase of maximum water temperature in summer, that can reach +8 °C in some studies (Cunningham *et al.*, 2023; Johnson & Jones, 2000).

For example, a study conducted in 2022 in British Columbia showed that streams had a maximum water temperature 3.7°C higher where 35% of the buffer strips were harvested since 1970, than where only 5% of the buffer strips were harvested (Cunningham *et al.*, 2023). Rutherford *et al.* (1997) showed that moderate shade levels (c. 70%) may be sufficient to restore stream temperatures in the thermal tolerance limits of

aquatic fauna (close to 20°C) in a small pasture under temperate climates. Ebersole et al. (2003) observed a 2-4°C decrease in the daily maximum temperature of cold-water plumes after the implementation of shading cover along the stream.

The optimal buffer width to preserve stream thermal quality and ecosystems varies from a site to another, but most of studies conclude to the following minimum values:

- 10 to 15 m to preserve stream physical and chemical characteristics such as temperature (e.g., Broadmeadow & Nisbet, 2004; Clinton, 2011; Dan Moore *et al.*, 2005; Gomi *et al.*, 2006; Sweeney & Newbold, 2014; Wilkerson *et al.*, 2006). However, emphasis should be placed on tree height and density to maximize stream shading. For instance, DeWalle (2010) showed that at least 80% shade on small streams up to 6-m wide can be achieved in mid-latitudes with relatively narrow 12-m wide buffers provided that buffers are tall and dense.
- 30 m to preserve hydrogeomorphological characteristics of streams as well as their ecological integrity, including benthic communities (Broadmeadow & Nisbet, 2004; Newbold *et al.*, 1980; Sweeney & Newbold, 2014).
- 50 m for an optimal protection of aquatic ecosystems, especially in erosion sensitive areas (Broadmeadow & Nisbet, 2004; Martel *et al.*, 2007).

Scientific literature is also unanimous in stating that the delineation of riparian zones should, as much as possible, follow a systematic approach and be based on ecosystemic and geomorphological characteristics, particularly the width of watercourses and their floodplains (Holmes & Goebel, 2011; Tormos, 2010). It should also consider local knowledge about soil susceptibility to erosion, slope effects on the banks, drainage quality, and the potential for windthrow (FQSA-DGR, 2012).

Several studies recommend riparian zones consisting of at least two areas: an integral protection zone near the watercourse, at least 10 m wide, and one or two outer, wider zones with partial harvesting, the width of which can be adapted based on site characteristics, including slope, soil type, and vegetation cover (Haberstock et al., 2000; Hawes & Smith, 2005; Kleinschmidt Associates, 1999). In the study by Haberstock et al. (2000), which was conducted with the specific goal of protecting salmon rivers in northeastern United States, the recommended width of the first integral protection zone is 11 m, while the second partial harvesting zone ranges from 12 to 80 m. The total absence of interventions near the watercourse aims to preserve the soil's absorption capacity as much as possible.

Most studies also emphasize the importance of preserving riparian zones of small headwater streams, which, at the watershed scale, have a greater influence on water temperature and water quality than buffer zones of larger downstream watercourses (Hawes & Smith, 2005). Additionally, these riparian zones often have poor drainage and are therefore fragile in terms of heavy machinery movement. This is particularly true for intermittent watercourses, which have the highest proportion of fragile riparian areas (Bertrand, 2007). Therefore, clearcutting should be avoided in intermittent stream riparian zones, and if partial harvest is done, it should be a way to maintain as far as possible buffer zone function (including shading) over time, as stated in the *Forest Management manual for New Brunswick Crown Lands* (DNRED, 2014).

3. Cold-water patches and thermal refuges

A cold-water patch (CWP) is generally defined as a discrete patch of water that is cooler than the surrounding ambient water (Mejia et al., 2023). CWPs vary in size, location and configuration over space and time due to complex riverscape processes including geomorphology, topography, geology, hydrology

and hydrogeology (Ebersole et al., 2015; Monk et al., 2013; Wawrzyniak et al., 2016). CWP can be classified into seven well-defined hydrological types: tributary confluence plume, lateral spring or seep, springbrook, side channel, alcove, wall-base channel and hyporheic upwelling (Torgersen et al., 2012). The CWPs linked to the drainage network (confluence plumes, springbrooks, side channels) are the most easily identifiable due to their predictable locations across the riverscape, while tributary confluence plumes often provide the highest volume of cold water, allowing the persistence of large numbers of thermoregulating cold-water fishes (Wang et al., 2020; Sullivan & Vokoun, 2023).

When a CWP is used by cold-water fishes to avoid higher temperatures during warm periods, it can then be identified as a cold-water refuge (CWR) (Sullivan et al., 2021; Mejia et al., 2023). CWPs and CWRs contribute to overall thermal heterogeneity and are important determinants of river-wide population dynamics for freshwater fishes (Mejia et al., 2023). In the context of climate change, river-wide thermal heterogeneity, and more specifically, the presence and quality of CWRs, is critical in determining fish persistence and survival in rivers, especially at lower latitudinal or elevational extents of their range (Corey et al., 2023; Wenger et al., 2011). Therefore, it is imperative that land management strategies include the specific protection of CWPs and CWRs (Mejia et al., 2023; Morelli et al., 2016; Reside et al., 2019). This is done through the definition of specific buffer zones with total protection, as it is the case in the *Forest Management manual for New Brunswick Crown Lands* (DNRED, 2014) with a 30 m buffer without any timber harvesting around “cold-water inputs and refuges”. In the Restigouche River watershed, this governmental database contains only 7 cold-water inputs and refuges, all located in the Kedgwick River and the Little Main Restigouche River, on Licence 9 territory. No such cold-water inputs are identified in the eastern part of the watershed (Licence 1 territory). According to DNRED, this identification was made several years ago by DNRED with help from regional biologists. In most cases, it was based on biologists’ knowledge of the territory. However, thermal anomalies in the Restigouche River watershed were identified and classified between 2011 and 2013 using high-resolution optical and thermal images, yielding a database of 1,825 classified thermal anomalies throughout the whole Restigouche River watershed (Dugdale et al., 2015). This includes more than 171 confluence plumes in the New Brunswick part of the watershed that could potentially be considered as “cold-water inputs and refuges”. This discrepancy questions the relevance of the current “cold-water inputs and refuges” governmental database and highlights the need for updating it to be able to adequately protect cold-water refuges and effectively inform decision-making with robust and reliable data.

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